

```
! pip install tf-keras
```

```
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```
from sentence_transformers import SentenceTransformer, util
model = SentenceTransformer("all-MiniLM-L6-v2")
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate li
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set

modules.json: 100%	349/349 [00:00<00:00, 31.8kB/s]
config_sentence_transformers.json: 100%	116/116 [00:00<00:00, 9.75kB/s]
README.md: 100%	10.5k/10.5k [00:00<00:00, 809kB/s]
sentence_bert_config.json: 100%	53.0/53.0 [00:00<00:00, 5.47kB/s]
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Loading weights: 100%	103/103 [00:00<00:00, 2629.95it/s]
tokenizer_config.json: 100%	350/350 [00:00<00:00, 30.2kB/s]
vocab.txt: 100%	232k/232k [00:00<00:00, 7.93MB/s]
tokenizer.json: 100%	466k/466k [00:00<00:00, 22.3MB/s]
special_tokens_map.json: 100%	112/112 [00:00<00:00, 10.6kB/s]
config.json: 100%	190/190 [00:00<00:00, 18.6kB/s]

```
s1 = "The is on the mat"
s2 = "A cat is sitting on the mat"
emb1 = model.encode(s1)
emb2 = model.encode(s2)
sim_score = util.cos_sim(emb1, emb2)
sim_score
```

```
tensor([[0.4999]])
```

```
sentences = [
    "The cat sits outside",
    "A man is playing guitar",
    "I love pasta",
    "The new movie is awesome",
    "The cat plays in the garden",
    "A woman watches TV",
    "The new movie is so great",
    "Do you like pizza?",
    "She enjoys reading books",
    "He is walking his dog",
    "The weather is nice today",
    "Let's go to the beach",
    "I need a cup of coffee",
    "They are planning a trip",
    "She cooks delicious meals",
    "I am learning to play the piano",
    "He works at a tech company",
    "This book is very interesting",
    "The sun is shining brightly",
    "It's raining outside",
    "The children are playing soccer",
    "He is fixing the car",
    "She writes beautiful poetry",
    "The flowers are blooming",
    "I enjoy hiking in the mountains",
    "He is swimming in the pool",
    "We are watching a comedy show",
    "The dog is barking loudly"
```

```
"The dog is barking loudly",  
"She is painting a landscape",  
"He likes to play video games",  
"I am studying for my exams",  
"They are building a new house",  
"The coffee shop is busy",  
"He listens to classical music",  
"She is designing a website",  
"I bought a new laptop",  
"They are baking cookies",  
"The bird sings in the morning",  
"He loves going to concerts",  
"She practices yoga every day",  
"The phone is ringing",  
"I need to finish this project",  
"The restaurant has great food",  
"He collects rare coins",  
.... "She enjoys watching documentaries",  
"They are exploring the city",  
"The laptop is charging",  
"He is training for a marathon",  
"I meditate every morning",  
"The bakery sells fresh bread",  
"She is learning French"  
]
```

```
paraphrases = util.paraphrase_mining(model, sentences)  
paraphrases
```



```
[0.031751833856105804, 6, 39],
[0.031715165823698044, 40, 49],
[0.03155869245529175, 47, 48],
[0.03062995709478855, 29, 36],
[0.03037250168022001, 35, 38]
```

```
for i_data in paraphrases[0:20]:
    score, idx1, idx2 = i_data
    print(f"{sentences[idx1]} \t\t\t {sentences[idx2]} \t\t\t Score : {score:.4f}")
```

The new movie is awesome	The new movie is so great	Score
The cat sits outside	The cat plays in the garden	Score : 0.6788
She enjoys reading books	She enjoys watching documentaries	
They are planning a trip	They are exploring the city	Score
He listens to classical music	He loves going to concerts	Score
I need a cup of coffee	The coffee shop is busy	Score : 0.5284
He likes to play video games	He loves going to concerts	Score
I bought a new laptop	The laptop is charging	Score : 0.5188
I love pasta	Do you like pizza?	Score : 0.5096
She enjoys reading books	She is learning French	Score
She cooks delicious meals	The restaurant has great food	Score
He likes to play video games	He listens to classical music	Score
I am learning to play the piano	I am studying for my exams	
She writes beautiful poetry	She is painting a landscape	Score
The weather is nice today	The sun is shining brightly	Score
A woman watches TV	She enjoys watching documentaries	Score
She cooks delicious meals	She writes beautiful poetry	Score
She is painting a landscape	She is designing a website	Score
They are baking cookies	The bakery sells fresh bread	Score
She enjoys reading books	She writes beautiful poetry	Score

Outlier Detaction

```
model = SentenceTransformer("distiluse-base-multilingual-cased-v1")
```

```

modules.json: 100%                               341/341 [00:00<00:00, 29.6kB/s]
config_sentence_transformers.json: 100%           122/122 [00:00<00:00, 6.09kB/s]
README.md: 100%                                   2.24k/2.24k [00:00<00:00, 71.7kB/s]
sentence_bert_config.json: 100%                   53.0/53.0 [00:00<00:00, 2.05kB/s]
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special_tokens_map.json: 100%                    112/112 [00:00<00:00, 7.98kB/s]
config.json: 100%                                190/190 [00:00<00:00, 8.18kB/s]
config.json: 100%                                114/114 [00:00<00:00, 7.03kB/s]
2_Dense/model.safetensors: 100%                  1.58M/1.58M [00:00<00:00, 5.81MB/s]

```

```

corpus1 = ["The stock market saw a big drop",
           "Shares declined sharply due to economic news.",
           "Money gets money in stock",
           "it may rain today",
           "He collects rare coins",
           "She enjoys watching documentaries,"]
corpus_emb=model.encode(corpus1)

```

```

# Compute pairwise cosine similarity
cos_similarity = util.pytorch_cos_sim(corpus_emb, corpus_emb)
avg_sim = cos_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()

```

```
outlier_index  
print(f"Outlier sentence : {corpus1[outlier_index]} (AGV_Similarity :{avg_sim[outlier_index]}")
```

Outlier sentence : She enjoys watching documentaries, (AGV_Similarity :0.2605765163898468)

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